

Task 1 Report:

Objective

The goal of this task was to build a high-quality, labeled dataset of real-world cyber-physical incidents by combining LLM-assisted data collection with human-in-the-loop validation. Each incident was annotated with a set of category codes representing the nature and impact of the attack .

Data Acquisition and Automation

Query Design

We prepared a list of 50 diverse cyber-physical incident queries that covered a range of infrastructure sectors (e.g., water treatment, energy grid, hospitals, disaster systems) and attack types (e.g., ransomware, social engineering, drone-based disruption). These queries aimed to simulate realistic and emerging threats.

Example queries:

- "Ransomware attack on utility systems"
- "Quantum computer breach on encrypted data"
- "Phishing campaign disrupts billing systems"

Automated Data Collection

We developed a Python module `tools.py` using the **LangChain** framework and **TavilySearchResults** tool to automate the following steps for each query:

- Perform a real-time web search.
- Retrieve and truncate article content, article name, article link.
- Use **Google Gemini 2.0 Flash /1.5 Flash** via LangChain to:
 - Generate a structured, paraphrased incident summary.
 - Predict a list of applicable category codes for each article.

The results are stored as JSON objects including:

- `article_title`
- `article_url`
- `incident_summary`
- `llm_suggestion`

The dataset was saved to `labeled_datasetx.json`.

One time we set model to gemini 2.0 and then gemini 1.5 to get two raw datasets with all details.

Manual Labeling and Human Review

Review Strategy

For each incident summary in both datasets:

- I analyzed the content and LLM-suggested categories.
- An **ordered list of category codes** is manually determined to reflect the **chronological or causal flow of impacts**.
- A free-text **justification** note was written to explain why certain categories were added, reordered, or removed in LLM suggestions.

Labeling Criteria

Manual labeling was guided by official category definitions:

- **DCA**: Digital Compromise or Access
- **OIA**: Operational Infrastructure Attack
- **BPA**: Biological/Physical/Health Impact
- **NDA**: Natural Disaster
- **EFA**: Economic/Financial Attack
- **SPI**: Socio-Political Influence
- **PSI**: Physical Security & Infrastructure Attacks
- **AAT**: Emerging and Advanced Technology Attacks

Structure of the final dataset after manual labelling

"article_title": "Ransomware hits city water plant",

"article_url": "https://example.com/ransomware-attack",

"incident_summary": "A ransomware attack disrupted operations at a city water plant...",

"category_flow(human labeled)": ["DCA", "OIA", "BPA"],

"llm_suggestion": ["DCA", "OIA"],

"Note": "While LLM correctly predicted digital compromise and operational disruption, it missed the biological impact (BPA) due to halted water treatment and contamination risks."

Challenges and Observations

- **Incomplete Inference**: Gemini frequently captured primary impacts (like digital compromise) but missed secondary consequences (e.g., health risks, financial impact).

- **Under-prediction of PSI and SPI:** Physical and socio-political impacts were under-represented unless very explicitly mentioned.
- **Ambiguity in Multi-Phase Incidents:** Incidents involving multi-step consequences (e.g., a phishing attack that leads to a shutdown of medical equipment) were often oversimplified.